# SEPARATEDNESS OF THE STRUCTURE MORPHISMS IS EVENTUAL

THE STACKS PROJECT AUTHORS

*Editorial note.* No omissions, hints or 'immediate' steps in the counted span.

*Allowed standard prerequisites (editorial).*

- Limit separated. Limits, Lemma lemma-limit-separated (same chapter)

*Source reference.* Original lemma 086Q: sources/stacks-chow/tags.txt, source lines 10400–10400. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X = \lim X_i$ be a directed limit of schemes over $S$ with affine transition morphisms. Assume*

(1) *$S$ quasi-separated,*
(2) *$X_i$ quasi-compact and quasi-separated,*
(3) *$X \to S$ separated.*

*Then $X_i \to S$ is separated for all $i$ large enough.*

*Proof.* Let $0 \in I$. Note that $I$ is nonempty as the limit is directed. As $X_0$ is quasi-compact we can find finitely many affine opens $U_1, \ldots, U_n \subset S$ such that $X_0 \to S$ maps into $U_1 \cup \ldots \cup U_n$. Denote $h_i : X_i \to S$ the structure morphism. It suffices to check that for some $i \geq 0$ the morphisms $h_i^{-1}(U_j) \to U_j$ are separated for $j = 1, \ldots, n$. Since $S$ is quasi-separated the morphisms $U_j \to S$ are quasi-compact. Hence $h_i^{-1}(U_j)$ is quasi-compact and quasi-separated. In this way we reduce to the case $S$ affine. In this case we have to show that $X_i$ is separated and we know that $X$ is separated. Thus the lemma follows from Lemma 086Q. □

*Date*: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.